



Data Science Study Plan (June 11 to October 5, 2025)

Study Time: 6 hours/day (Mon to Sat), Sunday off

Goal: Be ready to apply for top data science roles by October 2025

Phase 1: Foundations (Week 1–4)

Dates: June 11 – July 5

Focus Areas: - Python (Basics to Intermediate) - SQL (Basics + Joins + Aggregations) - Statistics & Probability

Weekly Breakdown: - Week 1: - Python basics (lists, dicts, functions, loops, OOP) - NumPy & Pandas hands-on - 1 small EDA project on a Kaggle dataset - Week 2: - SQL: SELECT, WHERE, GROUP BY, HAVING, JOINS - Mini SQL project using a sample dataset - Python: file I/O, error handling, comprehension - Week 3: - Stats: mean, median, std, probability basics - Distributions, sampling, CLT, hypothesis testing - SQL practice: subqueries, window functions - Week 4: - Combine: Python + SQL + Stats in a mini project - Project: Exploratory Data Analysis end-to-end - Resume draft + GitHub setup

Phase 2: Core Data Science & ML (Week 5–8)

Dates: July 7 – August 2

Focus Areas: - Machine Learning (Supervised & Unsupervised) - Model evaluation & deployment - Continue LeetCode/SQL practice

Weekly Breakdown: - Week 5: - ML Concepts: regression, classification - sklearn pipelines, metrics, preprocessing - Project: Titanic survival prediction - Week 6: - Unsupervised Learning: clustering, PCA - Project: Customer segmentation - Leetcode: 5 problems/day (Python + SQL) - Week 7: - Model tuning (GridSearchCV, cross-validation) - Evaluation: confusion matrix, ROC, AUC, etc. - Blog: write 1 post explaining your ML project - Week 8: - Streamlit app for 1 ML project - GitHub polish - Resume update

Phase 3: Advanced Topics & Projects (Week 9–12)

Dates: August 4 – August 30

Focus Areas: - Deep Learning basics (NNs, CNNs) - Time Series OR NLP - Portfolio projects

Weekly Breakdown: - Week 9: - Deep Learning theory: activation, loss, optimizers - TensorFlow/Keras or PyTorch basics - Small DL project (digit classification) - Week 10: - Choose: Time Series or NLP path - Time Series: ARIMA, FBProphet - NLP: text cleaning, TF-IDF, sentiment analysis - Week 11: - Project: Build final

capstone project (choose use-case) - Integrate ML + visualization + web app - Week 12: - Finalize project - Prepare portfolio write-up (LinkedIn + GitHub) - Start resume targeting for companies

Phase 4: Interview Prep + Applications (Week 13–16)

Dates: September 1 – October 5

Focus Areas: - DSA + SQL + ML theory revision - Mock Interviews + Resume Finalization - Job Application

Weekly Breakdown: - Week 13: - SQL practice sets (intermediate–advanced) - Python Leetcode (medium level) - ML Concepts summary notes - Week 14: - Mock interviews (online platforms or with peers) - Resume polish + cover letter drafts - Job board shortlisting - Week 15: - Apply to Google, Microsoft, other top companies - Behavioral interview prep - Kaggle competition attempt (optional) - Week 16: - Final review of portfolio + GitHub - Apply to internships + entry roles - Continue practice + track applications

Final Deliverables by October 5, 2025: - [] GitHub with 3–4 high-quality DS/ML projects - [] Resume tailored for DS roles - [] LeetCode: 150+ problems - [] SQL: strong intermediate level - [] Notion or personal website portfolio - [] Applications submitted to 20+ top companies

Let's go make this happen!